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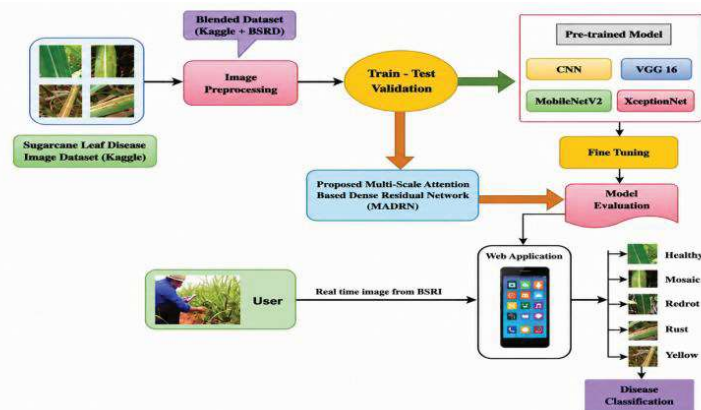
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accuracy and 95.00% AUC on the blended dataset, together with strong precision, recall, and F1-score values. These findings indicate that the proposed architecture provides robust and effective feature learning for complex sugarcane leaf disease patterns and maintains strong generalization under diverse environmental conditions. To support practical deployment, a web-based application was developed for real-time, user-friendly disease detection. The proposed framework offers a reliable and scalable solution for early disease diagnosis and decision support in sugarcane cultivation.



Keywords: Sugarcane; Leaf Disease; Deep Learning; Attention Mechanism; MADRN.

Paper ID: 108

Automated Five-Stage Diabetic Retinopathy Grading Using a Leak-Safe and Explainable Deep-Handcrafted Fusion Framework

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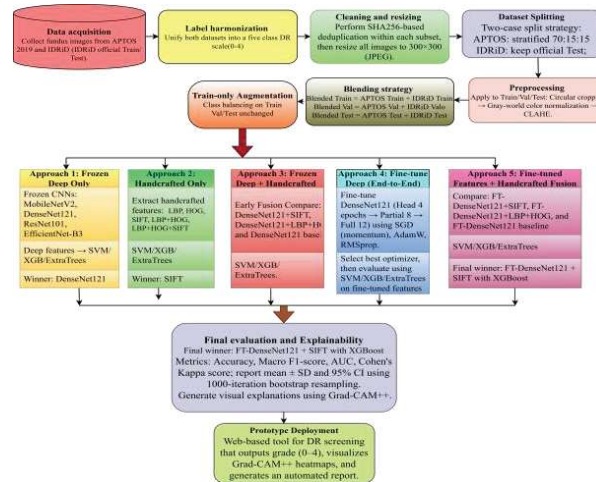
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Abstract:

Diabetic retinopathy (DR) is a major cause of preventable vision loss, yet fine-grained five-stage grading (Grades 0–4) from retinal fundus photographs remains expert-dependent and difficult to scale in resource-limited screening settings. This study proposes an explainable deep-handcrafted fusion framework for automated five-stage DR grading and introduces a unified leak-safe cross-source benchmark by harmonizing the APTOS 2019 and IDRiD datasets to support evaluation under more realistic mixed-source screening variability. To minimize data leakage, within-subset and cross-dataset duplicate screening was performed while preserving the official IDRiD test split. The resulting benchmark contains 4,045 unique images, including 3,534 from APTOS, 408 from IDRiD Train, and 103 from IDRiD Test. The proposed five-stage, validation-driven framework enables controlled comparison across frozen deep features, handcrafted descriptors, feature fusion, fine-tuning, and fine-tuned fusion under a consistent evaluation protocol. Performance was assessed on fixed Train/Validation/Test splits using Accuracy, Macro-F1, one-vs-rest AUC, and Cohen's kappa, with uncertainty estimated from 1,000 bootstrap resamples. DenseNet121 and SIFT-BoVW emerged as the strongest deep and handcrafted components, respectively. The final model, combining fine-tuned DenseNet121 embeddings, SIFT-BoVW, and XGBoost, achieved held-out test accuracy of $78.71\% \pm 1.65\%$, Macro-F1 of $64.13\% \pm 2.55\%$, Cohen's kappa of $67.88\% \pm 2.30\%$, and OvR AUC of $94.80\% \pm 0.59\%$. Compared with three lightweight custom CNN baselines trained end-to-end on the same inputs, the proposed fusion model delivered a stronger overall balance of accuracy, agreement, and class

separability. To support screening-oriented deployment, the selected model was further integrated into an explainable web-based prototype that reports predicted grade, class probabilities, and Grad-CAM++ visualizations. These findings highlight the potential of the proposed framework to support accurate, transparent, and scalable diabetic retinopathy screening in resource-limited settings.



Keywords: Diabetic Retinopathy; Retinal Fundus Images; Deep Learning; Handcrafted Features; Feature Fusion.

Paper ID: 115

Comparative Analysis of Missing Value Imputation Methods with a Hybrid Approach for HCV Prediction Using Machine Learning

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Abstract

Missing values in clinical datasets can substantially affect the reliability, stability, and diagnostic performance of machine learning models, particularly in small and imbalanced medical datasets such as Hepatitis C Virus (HCV) records. This study presents a comparative evaluation of missing-value imputation strategies for HCV prediction and disease-severity modeling using machine learning. The publicly available UCI HCV dataset was used, containing 615 samples and 12 demographic and biochemical predictors, including age, sex, albumin, alkaline phosphatase, alanine aminotransferase, aspartate aminotransferase, bilirubin, cholinesterase, cholesterol, creatinine, gamma-glutamyl transferase, and total protein. The dataset consisted of 540 healthy controls (87.8%) and 75 HCV-related cases (12.2%), representing hepatitis, fibrosis, and cirrhosis stages. Before model development, 31 natural missing values were identified, corresponding to approximately 0.36% missingness, mainly concentrated in ALP and CHOL. Four missing-value handling strategies, namely MICE, Median, Sampling-based imputation, and the proposed Hybrid MICE-Median approach, were evaluated under a leakage-safe analytical workflow using an 80:20 train-test split. Eight machine learning algorithms were assessed for binary HCV classification and disease-severity regression based on ordered HCV stages. Under complete-case analysis, 589 samples remained after excluding incomplete records, and XGBoost achieved the strongest regression performance with $R^2 = 0.827$, $MAE = 0.088$, and $RMSE = 0.270$, followed by Random Forest with $R^2 = 0.798$ and $RMSE = 0.288$. For binary HCV classification, XGBoost also produced the most balanced performance, achieving $MCC = 0.914$, $F1\text{-score} = 0.920$, $Accuracy = 0.984$, and $ROC\text{-AUC} = 0.998$. In the